

LECTURE 3: STATISTICAL DECISION THEORY

Michal Kolesár*

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1. BASIC CONCEPTS

REFERENCE Background reading: Chapter 1 in Ferguson (1967).

Statistical decision theory is an organizing framework that underlies everything we do in statistics, including estimation and hypothesis testing, but also many other things such as the design of experiments, designing sequential stopping rules (should we stop the experiment and report an estimate, or else wait and collect more data?), or multiple decision problems (among the teachers in this school district, which ones are underperforming?).

The task of a statistician is to come up with a *decision rule*: mapping from the data to a decision (what estimate to report, what list of underperforming teachers to provide, whether to reject the null hypothesis, whether to stop the experiment, etc.). We first outline the basic elements of this framework, and then discuss how the framework can be used to evaluate and compare different decision rules. The framework has deep roots—these are discussed in the digressions, since they require some solid micro theory background.

While statistical decision theory tells us how to *evaluate* different decision rules, it does not tell us how to *find* good ones. We'll spend much of the rest of the course discussing several common strategies, or *mechanisms* for generating decision rules, in the context of estimation, and in the context of testing.

Digression. Much of game theory originates with von Neumann and Morgenstern (1944). Abraham Wald then made the connection between their ideas and the theory of statistics, which he laid out in his landmark book (Wald 1950). The book laid the ground for statistical decision theory as we know it today. Indeed, according to Brown (2000), the term “statistical decision theory” is a condensation of Wald’s phrase “the theory of statistical decision functions”. This theory generalized the theory of tests and confidence intervals developed by Neyman and Pearson by showing that other types of problems in statistics (estimation, optimal stopping rules, etc.) can be treated similarly. ☒

A statistical decision problem can be thought of as a two-person game: statistician against nature. This game has 4 elements: A set Θ called the *parameter space*, an *action*

*Email: mkolesar@princeton.edu.

space $\mathcal{A}$ available to the statistician, a loss function $L: \mathcal{A} \times \Theta \rightarrow \mathbb{R}$, and $F_X(\cdot | \theta)$, the cumulative distribution function (CDF) of the data X . Nature chooses a parameter $\theta \in \Theta$, which the statistician doesn't see (θ is sometimes called state of nature), and the statistician chooses an action $a \in \mathcal{A}$. The loss function $L(a, \theta)$ reflects the loss (negative utility) to the statistician when they take action a and the parameter is θ . Ideally, we choose the loss function based on the economics of the problem; often, however, we choose it to qualitatively reflect what we are trying to do and to be mathematically convenient, analogous to the choice of the form of utility in economic theory.

We depart from classic game theory, in two ways. First, we don't make assumptions about nature's payoff function: nature, in particular, doesn't have to act rationally. Second, the statistician is allowed to gather some information about nature's action by observing data $X \sim F_X(x | \theta)$. The set of possible distributions $\{F_X(\cdot | \theta): \theta \in \Theta\}$ is called a *statistical model*. If Θ is finite-dimensional, the model is *parametric*, otherwise it is non-parametric. A *statistical decision problem* is the triple $(\Theta, \mathcal{A}, L)$ coupled with the data X . We denote the probability density function (PDF) of X by $f_X(x | \theta)$. For a fixed x , $f_X(x | \theta)$ as a function of θ is called the *likelihood function*.

Remark 1 (Random sample). Typically, the data distribution $F_X(\cdot | \theta)$ will have a particular structure, reflecting how the data was collected. The simplest and most common case is that X consists of multiple independent observations, each of which has the same distribution: $X = (X_1, \dots, X_n)$, with X_i independent and identically distributed (i.i.d.). We refer to this as having a *random sample*. That is, $\{X_i\}_{i=1}^n$ is a *random sample* of size n from distribution $F(\cdot | \theta)$ if $X_1, \dots, X_n$ are mutually independent and $X_i \sim F(\cdot | \theta)$ for each i . Then $F_X(x | \theta) = \prod_{i=1}^n F(x_i | \theta)$.

Thus, a random sample is an i.i.d. sample: the observations are independent, and they are drawn from the same distribution. If $X_1, \dots, X_n$ is a random sample from a distribution with PDF $f(x | \theta)$, then the joint PDF is, by independence, given by

$$f_X(x_1, \dots, x_n | \theta) = \prod_{i=1}^n f(x_i | \theta).$$

In spite of its ubiquity, the assumption that our dataset is a collection of i.i.d. samples is a bit mysterious—why is it a reasonable assumption? If you want to find out, read the optional Section 4 below that provides an elegant motivation for this assumption.

The statistician's job is to pick an action $a \in \mathcal{A}$ after observing $X = x$. If for each x in the sample space $\mathcal{X}$, this decision is deterministic, we can characterize it by the (non-randomized) *decision rule* $\delta(X)$, where $\delta: \mathcal{X} \rightarrow \mathcal{A}$. More generally, we may mix between two rules δ_1 and δ_2 , in which case the decision rule is random. For most decision problems, randomization is a useful technical device; however there are some important decision problems where the ability to randomize—to play a mixed strategy—is essential. Can you think of such situations?

The statistician has to commit to a decision rule before seeing X , at which point the loss is random. Therefore, the statistician evaluates the possible rule δ according to its

expected loss, or *risk*,

$$R(\delta, \theta) = E_\theta[L(\delta(X), \theta)],$$

where E_θ denotes expectation under the distribution $F_X(\cdot | \theta)$. The risk function represents the average loss when the true state of nature is θ and the statistician uses the rule $\delta(\cdot)$. The key point is that the choice of δ should depend only on the risk function.

Example 1 (Hypothesis testing). We are interested in testing the null hypothesis $\theta \in \Theta_0$, for some $\Theta_0 \subseteq \Theta$. Here the action space is $\mathcal{A} = \{\text{accept}, \text{reject}\}$. The loss is ℓ_I if we make a type I error (reject a true null), ℓ_{II} if we make a type II error (accept a false null), and zero otherwise. The risk is ℓ_I times the Type I error probability of the test if $\theta \in \Theta_0$, and it is ℓ_{II} times one minus the power of the test if $\theta \in \Theta_1$. The decision rule is called a *test*. We'll study this problem in detail later. $\square$

Example 2 (Estimation). We are interested in directly learning about nature's move, θ , so that $\mathcal{A} = \Theta$. The decision rule is called an *estimator*, and we usually use the notation $\hat{\theta}(X)$ rather than $\delta(X)$, often dropping the argument and just writing $\hat{\theta}$. The particular action $\hat{\theta}(x)$ that we take is called an *estimate*. If Θ has many dimensions, we're only typically interested in part of it. If $\theta = (\beta, \gamma)$ and we only want to estimate β and don't care about γ , then γ is called a *nuisance parameter*, and β is called the *parameter of interest*.

There are many possible loss functions that one can use for evaluating estimators. Suppose the parameter of interest β is a scalar. Then we can use the absolute value loss, $L(\hat{\beta}, \theta) = |\hat{\beta} - \beta|$, zero-one loss $L(\hat{\beta}, \theta) = \mathbb{1}\{|\hat{\beta} - \beta| > k\}$ where k is some constant, etc. However, the most common loss function for evaluating estimators is quadratic loss $L(\hat{\beta}, \theta) = (\hat{\beta} - \beta)^2$, the risk of which is called mean squared error (MSE):

$$\begin{aligned} \text{MSE}(\hat{\beta}, \theta) &= E_\theta[(\hat{\beta} - \beta)^2] \\ &= E_\theta[(\hat{\beta} - E_\theta[\hat{\beta}] + E_\theta[\hat{\beta}] - \beta)^2] \\ &= \text{var}_\theta(\hat{\beta}) + (E_\theta[\hat{\beta}] - \beta)^2 + 2E_\theta[\hat{\beta} - E_\theta[\hat{\beta}]](E_\theta[\hat{\beta}] - \beta) \\ &= (E_\theta[\hat{\beta}] - \beta)^2 + \text{var}_\theta(\hat{\beta}) \end{aligned}$$

The quantity $E_\theta[\hat{\beta}] - \beta$ is called the *bias* of the estimator, and therefore we have the MSE decomposition

$$\text{MSE} = \text{bias}^2 + \text{variance}.$$

Both the bias and the variance in general depend on θ . If the bias is zero for all θ , then the estimator is called *unbiased*. We'll spend the next few lectures on this decision problem. $\square$

There are several other types of decision problems that we will not have time to study. Here are two prominent examples:

Example 3 (Optimal treatment allocation). There are two treatments, 0 and 1, with average outcomes under each treatment given by μ_0 and μ_1 . Analogous to testing, there are two actions $\delta \in \{1, 0\}$, either recommend the treatment or not. However, in contrast to

testing, the loss is symmetric—we care about the average outcome under the decision:

$$L(\delta, \mu) = -\mu_0(1 - \delta) - \mu_1\delta = -\mu_0 - (\mu_1 - \mu_0)\delta.$$

Examples of papers dealing with versions of this problem are Manski (2004), Kitagawa and Tetenov (2018), or Athey and Wager (2021). $\square$

Example 4 (Best subset selection). Let θ denote a vector of teacher “value-added”, so that θ_j is the average gain in test scores students experience when assigned to teacher j . But θ_j may also be quality of neighborhood j , quality of doctor j , or bias of judge j . The decision rule δ is a vector of binary indicators, $\delta_j = 1$ means we select the teacher, while $\delta_j = 0$ means we do not. If we want to select teachers with value added exceeding some threshold c , a natural loss is $L(\delta, \theta) = -\sum_j (\theta_j - c)\delta_j$. There are many applications that are motivated by this type of problem. See, for instance, Bergman et al. (2024), Chetty and Hendren (2018), or Arnold, Dobbie, and Hull (2022). $\square$

Digression. We said that the choice of δ should depend only on the risk function. This raises two questions: why should we use expected loss, and how to interpret the probability measure P_θ (induced by $F_X(\cdot | \theta)$) used to calculate the expectation. The interpretation we use is that P_θ reflects “objective” probabilities or “common knowledge”: if we are told what θ is, then everyone agrees on the distribution of the data X —there is no ambiguity. Note this interpretation doesn’t require the notion of repeatedly sampling the data—only that we agree on P_θ , however each one of us interprets or motivates it. In contrast, there is non-probabilistic ambiguity about θ : we may not even agree that our beliefs about θ can be formulated in terms of a probability distribution, let alone that the distributional beliefs are the same for different people. This separates *risk* from *ambiguity*, and is helpful for formulating statistical models: all common beliefs about X are encoded into the statistical model $F_X(x | \theta)$, while the rest is captured by the model parameter θ .

Why use risk? The idea that we should pick among decision rules based on their risk properties alone rests on the von Neumann–Morgenstern expected utility theorem from microeconomic theory (Mas-Collel, Whinston, and Green 1995, Proposition 6.B.3): if, after being told θ (but before seeing the data) the statistician has preferences over decision rules that satisfy the independence and continuity axioms, then there exists a loss function L (negative utility function) such that those preferences can be rationalized by comparing the risk associated with the decisions: δ_1 is preferred to δ_2 iff $R(\delta_1, \theta) \leq R(\delta_2, \theta)$.

The reason to use von Neumann–Morgenstern decision theory is that it ensures coherence of our decisions, much like probability axioms ensure coherence when assigning probabilities. You can start out by specifying a loss and the model $F_X(x | \theta)$, and deduce decision recommendations, or you can go the other way around, deduce the loss function and the probabilities from decisions. What’s more, you can go back and forth, adjusting the loss, the probabilities, or the decisions until you iron out all inconsistencies and incoherence. $\square$

Ideally, we would like to construct a decision rule that is at least as good as all other decision rules for all θ , i.e. pick a rule $\delta(X)$ such that $R(\delta, \theta) \leq R(\delta', \theta)$ for all rules δ' . That is generally impossible. However, it’s reasonable to impose a much weaker requirement of *admissibility*.

Definition 1. A decision rule δ is called *inadmissible* if there exists another rule δ^* with uniformly smaller risk, $R(\delta^*, \theta) \leq R(\delta, \theta)$ for all θ , with the inequality being strict for at least one θ . A decision rule that is not inadmissible is *admissible*.

Note that admissibility is defined with respect to a specific loss function, although for certain problems there exist decision rules that are admissible for a large class of loss functions.

Example 5. Suppose that $X = (X_1, \dots, X_n)$ with $X_i \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d., and σ^2 known, so that $\theta = \mu$, and $\Theta = \mathbb{R}$. Then $\bar{X}_n$ is unbiased for μ , and has variance σ^2/n . Therefore,

$$\text{MSE}(\bar{X}_n, \mu) = \sigma^2/n.$$

Now consider estimating μ using the sample median, $X_{((n+1)/2)}$ (assuming n is odd for simplicity). It's also unbiased (by symmetry). Laplace showed that the variance of the sample median is approximately (for n large) equal to $\frac{1}{4nf(m)^2}$, where f is the PDF of X_i and m is the median of X_i . In our case, this implies

$$\text{var}(X_{((n+1)/2})) \approx \frac{\sigma^2}{4n\phi(0)^2} = \frac{2\pi\sigma^2}{4n} = \frac{\pi}{2} \frac{\sigma^2}{n} > \frac{\sigma^2}{n}.$$

Therefore, the sample median is not admissible (under quadratic loss).

On the other hand, the estimator $\hat{\mu} = 1$ (that disregards the data) is admissible, since no other estimator achieves exactly zero risk at $\mu = 1$. $\boxtimes$

Admissibility is a very weak requirement: for a typical estimation or testing problem, there will be a lot of admissible estimators or tests.

This raises the question of how to choose the “best” decision rule among the admissible ones. To help narrow down the choices, one option is to restrict the class of decision rules. The two most common ways of doing this are by requiring unbiasedness (discussed in Section 2 below), or by appealing to the invariance principle (which we won't discuss). But even then, the number of remaining options may still be large. How, therefore, should we in general rank two risk functions that cross? The two most common ways of doing this are:

1. Use some weight function $\pi(\theta)$ (that is non-negative and integrates to one over Θ) and rank the decision rules by their weighted average risk, called the *Bayes risk*

$$r(\delta, \pi) = \int_{\Theta} R(\delta, \theta) \pi(\theta) d\theta.$$

A decision rule that minimizes this risk is called *Bayes rule* (with respect to the weighting function $\pi(\theta)$).

2. Use the worst-case risk:

$$\bar{R}(\delta) = \sup_{\theta \in \Theta} R(\delta, \theta).$$

A decision rule that minimizes the worst-case risk is called *minimax*. This criterion was proposed by Wald (1950).¹

A compelling justification for using Bayes risk is that the weight function π reflects the subjective beliefs about θ by the statistician; they are called the statistician's *prior*.

¹ Wald did not offer much in the way of motivation, writing in Chapter 1: “a minimax solution seems, in general, to be a reasonable solution of the decision problem when an a priori distribution for θ does not exist or is unknown to the experimenter”.

Digression. This justification is compelling because of the Anscombe and Aumann (1963) subjective expected utility theorem. See Mas-Collel, Whinston, and Green (1995, Proposition 6.F.1) or Ferguson (1967, Section 1.4) for textbook versions of the result; the following version is based on Stoye (2012). Let $\succsim$ denote the statistician's preferences over risk functions, so we write $r_1 \succsim r_2$ if the risk function $r_1 = R(\delta_1, \theta)$ is preferred to $r_2 = R(\delta_2, \theta)$. Suppose $\succsim$ fulfills the independence of irrelevant alternatives (IIA) axiom (whether $r_1 \succsim r_2$ doesn't depend on whether $r_3 = R(\delta_3, \theta)$ is available) and independence ($r_1 \succsim r_2$ implies $\alpha r_1 + (1 - \alpha)r_3 \succsim \alpha r_2 + (1 - \alpha)r_3$ for any risk function r_3 and $\alpha \in (0, 1)$). Then there exists a distribution Π over θ such that $R(\delta_1, \theta) \succsim R(\delta_2, \theta)$ iff $\int R(\delta_1, \theta) d\Pi(\theta) \leq \int R(\delta_2, \theta) d\Pi(\theta)$. In other words, if the statistician's preferences are "reasonable" in that they satisfy the independence and IIA axioms, then there must be a prior distribution Π that rationalizes them. $\square$

The beauty of this approach is that, as we'll see in Lecture 7 when we discuss Bayes rules in detail, all the statistician has to do is to use the data to update their beliefs, forming a *posterior*. Then, whatever the decision problem, it is easy to calculate the Bayes rule.

This is a compelling justification if the statistician's task is to make a decision that *they* believe is the best. This is relevant if the statistician runs a hedge fund and the decision is whether to make some trade, or if they work for a company. But often, the statistician's goal is not to make a decision that they believe is the best, but to communicate to an audience, or to act on behalf of a group of people, who may have different beliefs. Who may such a group be? One idea is that "the group will in fact be a rather nebulous body of people, for example the group of all specialists in some field" Savage (1972, p. 175). Perhaps it is an FDA jury deciding whether to approve a drug, or the statistician is communicating their research findings to the research community.

This group agrees on the likelihood $f(x | \theta)$, but each member may have different beliefs π about θ (we can always make sure that the audience agrees on the likelihood by making the model sufficiently rich: in fact, that's what we do in research! See also digression about separating risk and ambiguity above).

In such scenarios, a convincing argument in favor of minimax rules instead of using Bayes rules was made by Savage (1972, Chapter 10), and goes as follows.² To make sure everybody in the group agrees on a common decision rule δ , we pick a rule that ensures that the expected loss, as perceived by each member, is acceptable: we pick a rule that minimizes the maximum risk,

$$\sup_{\pi \in \Gamma} r(\delta, \pi) = \bar{R}(\delta),$$

where Γ is the set of priors held by the audience, and the equality holds if Γ is sufficiently rich. To see this, note that if Γ contains mass points (degenerate priors that put all probability on one θ), then clearly $\sup_{\theta \in \Theta} R(\delta, \theta) \leq \sup_{\pi \in \Gamma} r(\delta, \pi)$. In the other direction, for any π , $\int R(\delta, \theta) \pi(\theta) d\theta \leq \int \sup_{\theta \in \Theta} R(\delta, \theta) \pi(\theta) d\theta = \bar{R}(\delta)$. This justifies the use of a minimax rule if our goal is to act on behalf of or to communicate to a diverse audience: if the statistician suggests using a minimax rule, then, before seeing the data, "no member will feel that the suggestion is a serious mistake." (Savage 1972, p. 174).

2. Savage later said that minimax rules are "ill-founded".

Key to Savage's group decision-making interpretation is that the group commits to the rule ex-ante. After the experiment has been run, some members of the group will have ex-post regret about accepting the decision rule. This ex ante commitment is important—otherwise the audience may worry about p -hacking etc. This is also the reason for submitting pre-analysis plans when running randomized experiments.

Digression. In contrast, always defending the minimax principle on purely frequentist grounds appears to be difficult, as argued in Savage (1972, Chapter 9.7). Suppose we have to decide whether to bet \$2 for or against the event that electric cars will entirely replace the internal combustion engine next year. The minimax solution is to bet for with probability $1/2$, and bet against with probability $1/2$; few people would find this reasonable. $\boxtimes$

Digression. There is another justification for using the minimax rule even in settings in which the statistician doesn't act on the behalf of others, namely that there is *ambiguity* about the right distribution for θ . One of the few things that Fisher and Neyman agreed upon was the statistician has seldom enough information about the unknown state of nature θ to build a prior distribution for it: effectively, they argued that the statistician's preferences over risk functions may not satisfy the independence axiom. If we weaken the independence axiom to c -dependence (that is the independence axiom holds only for r_3 that are constant risk functions) and impose ambiguity aversion (If $r_1 \succsim r_2$ and $r_2 \succsim r_1$, then $\lambda r_1 + (1 - \lambda)r_2 \succsim r_2$ for any $\lambda \in (0, 1)$; the intuition is that this type of randomization may constitute a hedge against ambiguous states), then Gilboa and Schmeidler (1989) have shown that there exists a class of priors Γ such that the statistician's preferences can be represented as minimizing the worst-case risk over Γ , $\sup_{\pi \in \Gamma} r(\delta, \pi)$. This is known as the Γ -minimax criterion, and reduces to the minimax criterion if Γ is rich enough. $\boxtimes$

Example 6. Have you ever disagreed with a friend about which restaurant to go to, and flipped a coin to resolve the issue? Then you employed a minimax decision rule. Drawing random samples, or randomly assigning individuals to treatment and control groups is also a minimax rule. $\boxtimes$

Example 5 (continued). MSE of the sample mean doesn't depend on μ , so that both the average and maximum risk of $\bar{X}_n$ is σ^2/n . Furthermore, we will show in Lecture 7 that $\bar{X}_n$ is minimax and admissible. This result generalizes to the multivariate case, provided we are still interested in a single mean. That is, we observe $X_i \sim \mathcal{N}(\theta, \Sigma)$, with $\theta \in \mathbb{R}^k$, Σ known, and the parameter of interest is θ_1 . Then the sample mean $\frac{1}{n} \sum_{i=1}^n X_{i1}$ is minimax and admissible under squared error loss, $L(\hat{\theta}_1, \theta) = (\theta_1 - \hat{\theta}_1)^2$.

Suppose we are interested in all the means, however, using the compound loss $L(\hat{\theta}, \theta) = \sum_{i=1}^k (\hat{\theta}_i - \theta_i)^2$. Suppose for simplicity $\Sigma = \sigma^2 I$. You may be tempted to still say that $\hat{\mu} = \bar{X}_n$ is minimax and admissible. This turns out to only be half-right, since while $\bar{X}_n$ is minimax, it's not admissible if $k \geq 3$ (Stein 1956). In particular, it's dominated by several shrinkage estimators, such as the James and Stein (1961) estimator

$$\hat{\mu}_{JS} = \left(1 - \frac{(k-2)\sigma^2/n}{\|\bar{X}_n\|^2}\right) \bar{X}_n,$$

where $\|x\| = \sqrt{\sum_i x_i^2}$ is the Euclidean norm. The James-Stein estimator thus pulls the sample mean towards zero. This is a surprising result (what's special about zero? What's the intuition for the inadmissibility of $\bar{X}_n$?), and it generated a new branch of statistical methods called "Empirical Bayes" methods. It also underlies many of the modern machine learning techniques, as well as large parts of nonparametric statistics. $\boxtimes$

Digression. The leading alternative to the Bayes and minimax criteria is minimax regret, which we mention here, but won't discuss further in this course. This criterion minimizes $\max_{\theta \in \Theta} \{R(\delta, \theta) - \min_{\delta'} R(\delta', \theta)\}$. Here $\min_{\delta'} R(\delta', \theta)$ corresponds to the risk of the oracle who knows θ . For estimation and testing problems, this risk is zero, and the minimax regret criterion is the same as the minimax criterion. But it can make a difference in other problems, like Example 3 or Example 4. For instance, consider the optimal treatment allocation problem, and suppose μ_0 is known; we only don't know the treatment effect $\tau = \mu_1 - \mu_0$. Then the minimax rule is to always stick to the status quo, since any other rule could lead to lower welfare if $\mu_1 - \mu_0 < 0$, even if the data has overwhelming evidence that the treatment effect is most likely positive. In contrast, if we observe data $\hat{\tau} \sim \mathcal{N}(\tau, \sigma^2)$, then the minimax regret rule is simply $\mathbb{1}\{\hat{\tau} > 0\}$. We can intuitively motivate the minimax regret criterion as the regret experienced by a decision maker who chose δ and then learned that parameter value θ obtained. The minimax regret criterion also has an axiomatic justification (see Stoye 2012). One disadvantage of it is that it is menu-dependent—adding another decision rule to the menu may affect the relative ranking of the previously available rules, violating the IIA axiom. Minimax regret also has the same group decision-making justification as minimax risk. $\square$

2. UNBIASED ESTIMATORS

The property of unbiasedness has an intuitive appeal, but it is a bit hard to justify. In simple problems, once we restrict attention to unbiased estimators, it is possible to find one that has minimal variance—much of old statistical theory is concerned with the question of finding such minimum variance unbiased estimators. The decision-theoretic justification for this is rather weak, as noted by Hodges and Lehmann (1950):

The principles most commonly applied in the selection of a point estimate are the principles of maximum likelihood and of minimum variance unbiased estimation. Both of these principles are intuitively appealing, but neither of them can be justified very well in a systematic development of statistics.

(I disagree with their assessment of maximum likelihood, for reasons we will explore later).

Example 7. Both the *sample mean*, $\bar{X}_n = \sum_{i=1}^n X_i/n$, and the *sample variance*, $S_n^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2/(n-1)$, are unbiased as estimators of the mean and variance, provided that $X_1, \dots, X_n$ is a random sample of size n . To show this, let μ denote the mean and σ^2 denote the variance. By linearity of expectation,

$$E[\bar{X}_n] = E\left[\sum_{i=1}^n X_i/n\right] = \sum_{i=1}^n E[X_i]/n = \sum_{i=1}^n \mu/n = \mu.$$

To show unbiasedness of S_n^2 , let $Y_i = X_i - \mu$ and $\bar{Y}_n = \sum_{i=1}^n Y_i/n = \bar{X}_n - \mu$. Note that $E[Y_i] = 0$. Thus, $E[Y_i^2] = \text{var}(Y_i) = \text{var}(X_i) = \sigma^2$. Since Y_i are independent,

$\text{var}(\bar{Y}_n) = \sigma^2/n$. Then

$$\begin{aligned} E[S_n^2] &= E \left[\sum_{i=1}^n (X_i - \bar{X}_n)^2 / (n-1) \right] = \sum_{i=1}^n E \left[\frac{((X_i - \mu) - (\bar{X}_n - \mu))^2}{n-1} \right] \\ &= \sum_{i=1}^n E \left[\frac{(Y_i - \bar{Y}_n)^2}{n-1} \right] = \sum_{i=1}^n \frac{E[Y_i^2] - 2E[Y_i \bar{Y}_n] + E[\bar{Y}_n^2]}{n-1} \\ &= \sum_{i=1}^n \frac{\sigma^2 - 2\sigma^2/n + \sigma^2/n}{n-1} = \sigma^2, \end{aligned}$$

as claimed. $\square$

When the distribution of the data is normal, we can actually say a bit more about the distribution of these statistics. We state the result here now for later use:

Theorem 1 (Casella and Berger 2002, Theorem 5.3.1). If $X_1, \dots, X_n$ are i.i.d. random variables with $N(\mu, \sigma^2)$ distribution, then

- (i) $\bar{X}_n$ and S_n^2 are independent;
- (ii) $\bar{X}_n \sim N(\mu, \sigma^2/n)$; and
- (iii) $(n-1)S_n^2/\sigma^2 \sim \chi^2(n-1)$.

Proof. Let $Y_i = X_i - \bar{X}_n$. Then the random variables $\bar{X}_n, Y_1, \dots, Y_n$ are jointly normal (why?). Simple calculation shows that $E[\bar{X}_n] = \mu$ and that $\text{var}(\bar{X}_n) = \sigma^2/n$, which implies part (ii). To show part (i), observe that for any $j = 1, \dots, n$,

$$\begin{aligned} \text{cov}(\bar{X}_n, Y_j) &= \text{cov}(\bar{X}_n, X_j - \bar{X}_n) \\ &= \text{cov}(\bar{X}_n, X_j) - \text{var}(\bar{X}_n) = \frac{1}{n} \sum_{i=1}^n \text{cov}(X_i, X_j) - \sigma^2/n \\ &= \sigma^2/n - \sigma^2/n = 0. \end{aligned}$$

Since uncorrelated jointly normal random variables are independent, we conclude that $\bar{X}_n$ is independent of $Y_1, Y_2, \dots, Y_n$. Moreover, since $S_n^2 = \sum_{i=1}^n Y_i^2 / (n-1)$, part (i) holds since functions of independent random variables are independent as well.

The proof of part (iii) is left as a homework exercise. $\square$

Digression. The statistic $T = (\bar{X}_n - \mu) / (S_n / \sqrt{n})$ is called the *t-statistic*. Using Theorem 1,

$$T = \frac{\bar{X}_n - \mu}{S_n / \sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma / \sqrt{n}} \frac{1}{\sqrt{S_n^2 / \sigma^2}} \sim \frac{\mathcal{N}(0, 1)}{\sqrt{\chi^2(n-1) / (n-1)}} \sim t(n-1)$$

since the $\mathcal{N}(0, 1)$ and $\chi^2(n-1)$ random variables are independent. Thus, we proved that if $X_1, \dots, X_n$ is a random sample from $\mathcal{N}(\mu, \sigma^2)$, then the *t-statistic* has *t-distribution* with $n-1$ degrees of freedom. $\square$

While in simple settings, unbiasedness leads to intuitive estimators, sometimes unbiasedness restricts the class of estimators too much in the sense that we're left with an empty set: no unbiased estimators exist.

Example 8 (Lehmann and Casella 1998, Example 2.1.2). Let $X_1, \dots, X_n$ be a random sample from a Bernoulli distribution with parameter p . Suppose that our parameter of interest is $\theta = 1/p$. Let $\hat{\theta}(X)$ be some estimator. Then

$$E[\hat{\theta}(X)] = \sum_{(x_1, \dots, x_n) \in \{0,1\}^n} \hat{\theta}(x_1, \dots, x_n) P((X_1, \dots, X_n) = (x_1, \dots, x_n)).$$

We know that for any $(x_1, \dots, x_n) \in \{0,1\}^n$,

$$P((X_1, \dots, X_n) = (x_1, \dots, x_n)) = p^{\sum_{i=1}^n x_i} (1-p)^{\sum_{i=1}^n (1-x_i)},$$

which is a polynomial of degree n in p . Therefore, $E[\hat{\theta}]$ is a polynomial of degree at most n in p . However, $1/p$ is not a polynomial at all. Hence, there are no unbiased estimators in this case. $\square$

In some other cases, unbiased estimators may be quite weird.

Example 9. Suppose there is an infinite number of independent trials, $X_1, X_2, \dots$ from Bernoulli distribution with parameter p . Suppose that $X_i = 0$ means failure in the i th trial while $X_i = 1$ means success. Suppose that instead of observing the sequence $\{X_i\}_{i=1}^\infty$, we observe only the number of failures before the first success. Denote the number of failures by Y . Then $P(Y = y) = (1-p)^y p$. Suppose that the parameter of interest is p . Let $\hat{p}(Y)$ be some estimator. Then $E[\hat{p}] = \sum_{y=0}^\infty \hat{p}(y) (1-p)^y p$. If $\hat{p}$ is unbiased for p , then $\sum_{y=0}^\infty \hat{p}(y) (1-p)^y p = p$. Equivalently, for all $0 \leq p < 1$,

$$\sum_{y=0}^\infty \hat{p}(y) (1-p)^y = 1.$$

Thus, the only unbiased estimator is $\hat{p}(0) = 1$ and $\hat{p}(y) = 0$ for all $y \geq 1$. Does it seem reasonable? $\square$

Often there is a trade-off between bias and variance: in the previous example requiring unbiasedness led us to an estimator with variance that was unreasonably large. Thus, we may prefer a slightly biased estimator to an unbiased one if the former has much smaller variance in comparison to the latter one (with the trade-off measured by MSE).

Example 10 (Basu's elephants, Basu 1971). A circus owner is shipping their 100 elephants, and needs to know their total weight. Weighing all 100 is too cumbersome, so the owner wants to weigh just one elephant, Sambo, who is known to be about average weight, and multiply the number by 100. The circus statistician, horrified that this procedure can't produce an unbiased estimate of the weight of the herd, devises an alternative plan: select Sambo with probability 99/100, and select each other elephant with probability 1/9,900. The statistician argues: "I recently read an article in the *Annals of Mathematical Statistics*, where it is proved that the Horvitz–Thompson estimator is the unique hyperadmissible estimator in the class of all generalized polynomial unbiased estimators". In this case, the estimator is 100/99 times Sambo's weight, if Sambo is

selected, since it's the inverse of Sambo's sampling probability. For all other elephants, including Jumbo, the largest elephant, the estimate would be 9,900 times the elephant's weight. That is how the statistician lost his circus job. $\boxtimes$

This example may seem a bit ridiculous, but I encourage you to revisit it when you learn about using inverse probability weighting to estimate average treatment effects.

3. ASYMPTOTIC PROPERTIES OF ESTIMATORS

Sometimes, the finite-sample risk of an estimator is hard to figure out, in which case we may want to pick an estimator based on its *asymptotic risk*. As a first step towards figuring out its asymptotic risk, in most problems, we start out by establishing the following two properties:

1. An estimator $\hat{\theta}$ is *consistent* for θ if $\hat{\theta} \xrightarrow{p} \theta$ as $n \rightarrow \infty$.
2. An estimator is *asymptotically normal* with asymptotic variance Σ if there exist sequences γ_n and a_n and a matrix Σ such that $a_n(\hat{\theta} - \gamma_n) \Rightarrow \mathcal{N}(0, \Sigma)$. a_n is called the *rate of convergence* of $\hat{\theta}$, and Σ its asymptotic variance. Often, we may be able to set $\gamma_n = \theta$ and $a_n = n^{1/2}$, in which case the estimator is called *root- n consistent*.

Remark 2. Note that Σ is not necessarily equal to the limit of the finite-sample variances of $\hat{\theta}$, $\lim_{n \rightarrow \infty} \text{var}(a_n(\hat{\theta} - \gamma_n))$. In fact, the limit might not even exist, but if it does, then Σ is necessarily smaller than or equal to the limit of the variances (by Fatou's lemma from measure theory): in general, if $X_n \Rightarrow X$, then $\liminf_{n \rightarrow \infty} \text{var}(X_n) \geq \text{var}(X)$.³ For instance in an instrumental variables model, the first moment of the instrumental variables estimator doesn't exist for any finite n on the exactly identified case, but the estimator is asymptotically normal.

Digression (Bickel and Doksum 1977, Exercise 4.4.9, p. 150). An example in which the inequality is strict: Let $X \sim \mathcal{U}[0, 1]$, and let $X_n = X$ if $X \leq 1 - 1/n$, and $X_n = a_n$ if $X > 1 - 1/n$. Then, as an exercise, show that for any sequence a_n , $X_n \Rightarrow X$, but if $a_n/\sqrt{n} \rightarrow \infty$ (e.g. $a_n = n$), then $\text{var}(X_n) \rightarrow \infty$, and if $a_n/\sqrt{n} \rightarrow a$, then $\text{var}(X_n) \rightarrow a^2 + 1/12$, while $\text{var}(X) = 1/12$. $\boxtimes$

Digression. Because of this inequality, it is also generally not a good idea to estimate the asymptotic variance via the bootstrap. More on this in 539B...

If an estimator is consistent, then, loosely speaking, the risk under squared error should converge to zero. Due to the caveat we just discussed that convergence in probability or distribution doesn't imply convergence of moments, to tighten up our loose speech, we would need to consider "trimmed" loss, like $\min\{(\hat{\theta} - \theta)^2, K_n\}$, with the trimming cutoff K_n possibly increasing with n . Then it is indeed true that consistency implies that the trimmed MSE converges to zero.

3. See, for instance, Lemma 3 in Hahn and Liao (2021) for a careful proof. The reason one needs to be a bit careful is that while $\liminf_{n \rightarrow \infty} E[X_n^2] \geq E[X^2]$ follows directly by Fatou's lemma, the result is not immediate if we center the random variables.

Continuing this train of thought, if an estimator is asymptotically normal, this means that we can approximate the (trimmed) MSE in large samples by the asymptotic MSE, $E[(\hat{\theta} - \theta)^2] \approx \Sigma/a_n^2 + (\gamma_n - \theta)^2$.

Example 11. Let $X_1, \dots, X_n$ be a random sample from some distribution with mean μ and variance σ^2 . Let $\bar{X}_n$ be our estimator of μ and $S_n^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2 / (n-1)$ be our estimator of σ^2 . If we add and subtract μ , we get

$$\begin{aligned} S_n^2 &= \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 = \frac{1}{n-1} \sum_{i=1}^n ((X_i - \mu) + (\mu - \bar{X}_n))^2 \\ &= \frac{1}{n-1} \sum_{i=1}^n (X_i - \mu)^2 + \frac{2(\mu - \bar{X}_n)}{n-1} \sum_{i=1}^n (X_i - \mu) + \frac{1}{n-1} \sum_{i=1}^n (\mu - \bar{X}_n)^2 \\ &= \frac{n}{n-1} \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - \frac{2n(\mu - \bar{X}_n)^2}{n-1} + \frac{n}{n-1} (\mu - \bar{X}_n)^2 \\ &= \frac{n}{n-1} \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - \frac{n}{n-1} (\mu - \bar{X}_n)^2. \end{aligned}$$

By law of large numbers (LLN), $\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 \xrightarrow{p} E[(X_i - \mu)^2] = \sigma^2$, and $\bar{X}_n \xrightarrow{p} \mu$. Also, $n/(n-1) \rightarrow 1$. Therefore, by the continuous mapping theorem (CMT), the whole expression converges in probability to $1 \cdot \sigma^2 - 1 \cdot (\mu - \mu)^2 = \sigma^2$. So $\bar{X}_n$ and S_n^2 are consistent for μ and σ^2 , respectively.

Are they asymptotically normal? By the central limit theorem (CLT), $\sqrt{n}(\bar{X}_n - \mu) \Rightarrow N(0, \sigma^2)$. As for S_n^2 , it follows from the last line of the preceding display that

$$\sqrt{n}(S_n^2 - \sigma^2) = \frac{n}{n-1} \left[\frac{1}{\sqrt{n}} \sum_{i=1}^n ((X_i - \mu)^2 - \sigma^2) - \sqrt{n}(\bar{X}_n - \mu)^2 \right] + \frac{\sqrt{n}}{n-1} \sigma^2.$$

By the central limit theorem,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n ((X_i - \mu)^2 - \sigma^2) \Rightarrow \mathcal{N}(0, \tau^2),$$

with $\tau^2 = E[((X_i - \mu)^2 - \sigma^2)^2]$. Note that $\tau^2 = \mu_4 - 2\sigma^2 E[(X_i - \mu)^2] + \sigma^4 = \mu_4 - \sigma^4$ with $\mu_4 = E[(X_i - \mu)^4]$. By Slutsky's theorem and CMT,

$$\sqrt{n}(\bar{X}_n - \mu)^2 = \frac{1}{\sqrt{n}} (\sqrt{n}(\bar{X}_n - \mu))^2 \xrightarrow{p} 0.$$

Finally, $(\sqrt{n}/(n-1))\sigma^2 \rightarrow_p 0$. So, by Slutsky's theorem again,

$$\sqrt{n}(S_n^2 - \sigma^2) \Rightarrow N(0, \tau^2).$$

At home, try to derive this result by considering the asymptotic distribution of the vector $n^{-1} \sum_i (X_i, X_i^2)'$ and applying the delta method. $\square$

Example 12. If we have a random sample $X_1, \dots, X_n$ of size n , the CDF of the distribution that puts mass $1/n$ at each data point X_i is called the *empirical CDF*, usually denoted $\hat{F}_n$. Thus, by definition,

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{X_i \leq x\},$$

where $\mathbb{1}\{\cdot\}$ is the indicator function, i.e. the function which equals 1 if its argument is true, and 0 otherwise. In other words, $\hat{F}_n(x)$ shows the fraction of observations that are smaller than or equal to x . The empirical CDF is unbiased and consistent:

Lemma 2. If we have a random sample $X_1, \dots, X_n$ of size n from a distribution with CDF F , then for any $x \in \mathbb{R}$, $E[\hat{F}_n(x)] = F(x)$ and $\text{var}(\hat{F}_n(x)) \rightarrow 0$ as $n \rightarrow \infty$. As a consequence, $\hat{F}_n(x) \rightarrow_p F(x)$ as $n \rightarrow \infty$.

Proof. Note that $\mathbb{1}\{X_i \leq x\}$ equals 1 with probability $P(X \leq x)$ and 0 otherwise. Thus, $E[\mathbb{1}\{X_i \leq x\}] = P(X \leq x) = F(x)$. Hence, $E[\hat{F}_n(x)] = F(x)$ by linearity of expectation. In addition, $\text{var}(\mathbb{1}\{X_i \leq x\}) = F(x)(1 - F(x))$ by the formula for variance of a Bernoulli random variable with parameter $p = F(x)$. Therefore,

$$\text{var}(\hat{F}_n(x)) = \sum_{i=1}^n \text{var}(\mathbb{1}\{X_i \leq x\}) / n^2 = F(x)(1 - F(x)) / n \rightarrow 0. \quad \square$$

Because $E[\mathbb{1}\{X_i \leq x\}] = F(x)$, we could alternatively have used the law of large numbers, from which the remaining claims follow directly. In particular, the strong law implies that $\hat{F}_n(x) - F(x) \xrightarrow{\text{a.s.}} 0$. Actually, an even stronger result holds:

Theorem 3 (Glivenko-Cantelli). If $X_1, \dots, X_n$ is a random sample from a distribution with CDF F , then

$$\sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F(x)| \xrightarrow{\text{a.s.}} 0.$$

Proof. See, e.g., Theorem 2.4.7 in Durrett (2019). $\square$

This is a remarkable result: notice we have assumed *nothing* about F , and yet the theorem says that in a very strong sense, the empirical CDF will be uniformly close to it in large samples. $\boxtimes$

4. RANDOM SAMPLING (OPTIONAL)

REFERENCE Background reading: Chapter 1 in Schervish (1995).

An elegant way of motivating the random sampling framework is through the concept of exchangeability.

Definition 2. A sequence of random variables $\{X_i\}_{i=1}^\infty$ is *exchangeable* if for any n and m , the joint distribution of $(X_n, \dots, X_{n+m})$ remains unchanged under permutations.

This just says that the labels i that we put on the observations don't matter. We're not taking any stance on relative frequencies or independence.

Theorem 4 (Schervish 1995, Theorem 1.47). An infinite sequence $\{X_i\}_{i=1}^\infty$ of Bernoulli random variables is exchangeable iff there is a unique distribution $\Pi(\theta)$ on $[0, 1]$ such that

$$P(X_1 = x_1, \dots, X_n = x_n) = \int \prod_{i=1}^n P_\theta(x_i) d\Pi(\theta),$$

where $P_\theta(1) = \theta$ and $P_\theta(0) = 1 - \theta$. Furthermore, if the sequence is exchangeable, the distribution $\Pi(\theta)$ is unique, and $\bar{X}_n - \theta \xrightarrow{a.s.} 0$ (this second part is known as de Finetti's strong law of large numbers).

So an infinite sequence is exchangeable if it's conditionally i.i.d. given *something*⁴. This is one of only two ways you can generate exchangeable random variables. The only other way: a *finite* sequence is exchangeable iff they are like draws from an urn without replacement. This is a very powerful result.

What's the implication of this result? If people believe that a random sequence is exchangeable (the labels don't matter), then they all believe that there exists a random variable θ such that conditional on it, the random variables are i.i.d. Bernoulli with parameter θ . Furthermore, θ is the long-run frequency of a success. We now see how statistical models emerge: more generally, if we have exchangeable data $\{X_i\}$ (not necessarily Bernoulli), then there is a random variable θ , such that conditional on it, the variables are i.i.d., with distribution F_θ (in the general case, we can think of θ as a one-to-one function of the limiting probability measure F). Note also that Π here can be interpreted as a prior distribution (that we may or may not want to take a stance on, as per the discussion above about Bayes vs minimax rules).

So De Finetti's representation theorem is central to motivating statistical models, even if it's never mentioned once we write one down.

Digression. Why do we need an infinite sequence in de Finetti's theorem? Suppose $P(X_1 = 0, X_2 = 1) = P(X_1 = 1, X_2 = 0) = 1/2$, while $P(X_1 = 1, X_2 = 1) = P(X_1 = 0, X_2 = 0) = 0$. These random variables are exchangeable. But the representation doesn't hold, since it would imply $0 = \int_0^1 \theta^2 d\Pi(\theta) = \int_0^1 (1 - \theta)^2 d\Pi(\theta)$, which implies that Π puts mass 1 on both 0 and 1, which is impossible. To get around this, we may assume that the sequence has length N , but that we look only at a small fraction $n \ll N$ of it—then the theorem is “almost” true, see Diaconis and Freedman (1980). $\square$

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4. For general non-Bernoulli random variables the theorem says (see Theorem 1.49 Schervish 1995) that the sequence is exchangeable iff it's i.i.d. conditional on a random probability measure. Furthermore, this random probability measure is the almost sure limit of the empirical CDF $\hat{F}_n$ (see Example 12). You may wonder what we mean by a random probability measure and by saying that it converges—we'll leave those things to an advanced probability class.

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